

Retail Pricing Optimization Analysis

Phase 1: Excel Analysis

1. About This Project

I used the Kaggle Superstore US dataset for this project.

The goal of this project is to check how discount affects profit and find out after what discount percentage the company starts losing money.

This is the first phase of my full data analysis project using:

- Excel
- SQL
- Python
- Power BI

This phase is only based on Excel analysis.

2. What I Did First

The original dataset was in one sheet. I separated it into 4 sheets:

Customer Sheet

Contains customer details

Orders Sheet

Contains order information

Product Sheet

Contains product details

Transaction Sheet

Contains sales, profit, quantity and discount, I separated these sheets to make the dataset look like a real database.

3. Data Cleaning

Before analysis, I fixed some issues.

Fixed Date Errors

Order Date and Ship Date were showing incorrect values. I corrected them.

Removed Duplicate Rows

I removed repeated records.

Checked Data

I checked if all important values were correct.

4. New Columns I Created

Discount Band

I grouped discounts into:

- 0%
- 1–10%
- 11–20%
- 21–30%
- 31–50%
- 50%+

This helped me compare profit at different discount levels.

Profit Margin

Formula used:

Profit ÷ Sales

This helped me understand how much profit was earned from each sale.

5. Pivot Table Analysis

Pivot 1: Discount Band vs Average Profit

Why I Made This Pivot

To check how profit changes when discount increases.

What I Found

- Low discount (1–10%) gave the highest average profit
- Profit started decreasing after 10%
- Profit became negative at 21–30%
- Higher discounts caused bigger losses

Insight

Giving small discounts helps sales and profit.

Giving large discounts causes loss.

Recommendation

The company should keep normal discounts below 20%.

Pivot 2: Category vs Discount Band

Why I Made This Pivot

To check if discount affects some product categories more than others.

What I Found

All categories became negative after 21–30% discount:

- Furniture
- Office Supplies
- Technology

Insight

This is not a problem in one category.

The discount strategy is affecting the whole business.

Recommendation

The company should apply better discount control across all product categories.

Pivot 3: Region vs Discount Band

Why I Made This Pivot

To check whether discount impact changes in different regions.

What I Found

Most regions showed lower profit when discount increased.

Very high discounts caused losses in almost all regions.

Insight

High discounting is risky in every region.

Recommendation

Different regions should be monitored carefully before giving large discounts.

6. Main Finding

The biggest finding from this analysis is:

Profit becomes negative after 20% discount

This means 20% is the important break point.

After this level, discounts start hurting business profit.

7. Final Recommendations

Based on all pivot table analysis:

Recommendation 1

Keep standard discounts below 20%

Recommendation 2

Discounts above 20% should need approval

Recommendation 3

Discounts above 30% should only be used for:

- Stock clearance
- Seasonal sales
- Special campaigns

Recommendation 4

Track region performance before giving heavy discounts

8. Final Conclusion

From Excel analysis, I found that discount has a strong effect on profit.

Small discounts help increase profit.

High discounts reduce profit and create losses.

The analysis shows that the company should control discounting and avoid giving discounts above 20%.

9. Tools Used

- Microsoft Excel
- Pivot Tables

- Lookup Functions
- Calculated Columns

10. Next Step

In Phase 2, I will use SQL to check these results using:

- CASE WHEN
- CTE
- Window Functions
- Joins

This will help confirm the findings from Excel.